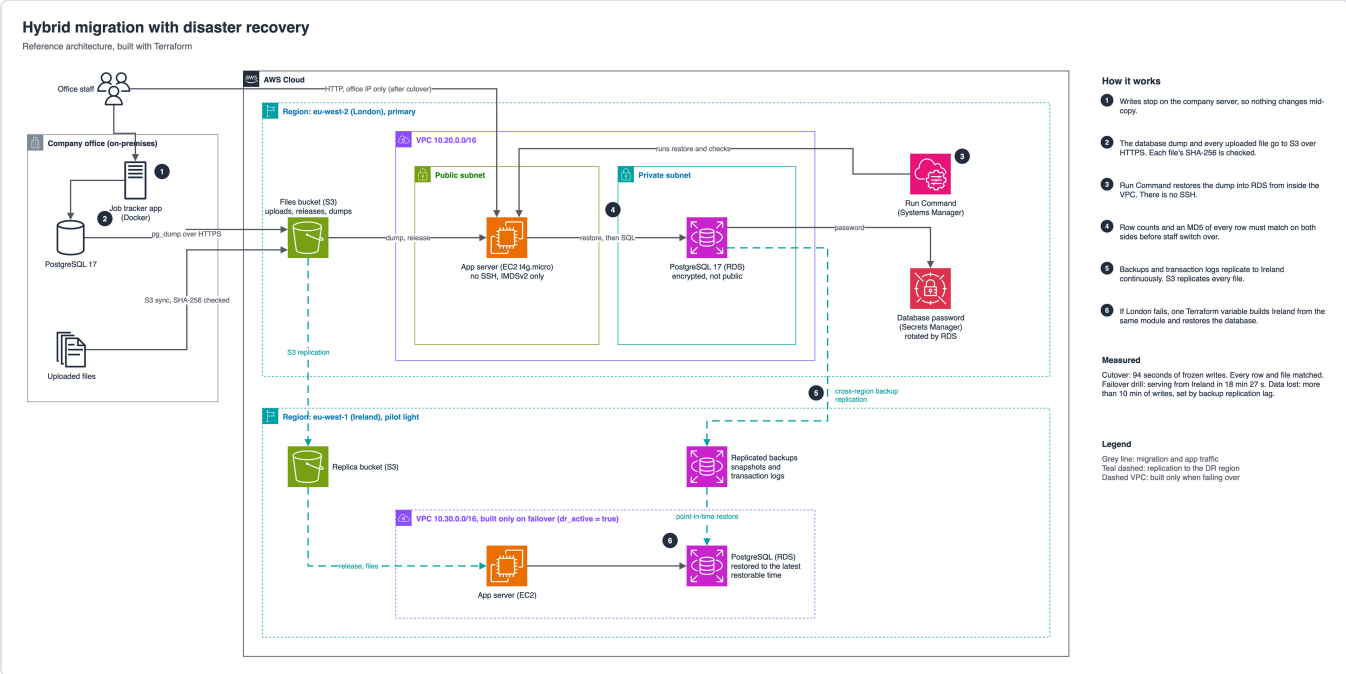
Hybrid migration with disaster recovery

Walkthrough in five parts: what each piece does, why it was built that way, and how to talk about it in an interview.

Repo: github.com/gumagatsby/aws-hybrid-migration-dr



The one-sentence pitch. I moved a small company's app, database and files from an office server to AWS in London, proved every row and file arrived unchanged, and then took London down on purpose to measure how fast the business recovers in Ireland and how much data it loses.

Part 1. The starting point and the target

The "company server" is a Docker Compose stack: a Flask job-tracking app, PostgreSQL 17 and a folder of uploaded files, seeded with 500 jobs and 60 attachments (plus one added by hand).

Piece	In the office	On AWS (London)	Why
App	Container on one server	EC2 `t4g.micro`, no SSH, IMDSv2 only	Rehost: the smallest change that gets it off the office box
Database	PostgreSQL in a container	Amazon RDS for PostgreSQL 17, private and encrypted	Re-platform: AWS takes over backups, patching and cross-region copies
Files	Local folder	S3, versioned, encrypted, TLS only	Durable, and replicable to another region
Password	Config file	Generated by RDS, held and rotated in Secrets Manager	Nothing secret in code or on disk

The app needed one change: read its database password from Secrets Manager and its files from S3, chosen by environment variables.

Part 2. The migration

`scripts/migrate.sh` runs five steps:

1. **Freeze writes.** Stop the app on the company server so nothing changes mid-copy.
2. **Dump the database** with `pg_dump` and upload it to S3 over HTTPS.
3. **Copy the files** with `aws s3 sync`, asking S3 to store a SHA-256 checksum of each file.
4. **Restore inside the VPC.** Systems Manager Run Command tells the app server to pull the dump and load it into RDS. There is no SSH and no open port 22.
5. **Validate.** Compare a row count and an MD5 of every row on both sides (time zone fixed so both servers print rows identically), and compare every file's local SHA-256 with the checksum S3 stored.

Measured	Result
Database	501 rows, identical hash on both sides
Files	61 of 61 SHA-256 matches
Downtime	94 seconds from freezing writes to validated cutover

A bug the validation caught in my own script. The first run reported a database mismatch. The data was fine: the AWS CLI adds a blank line to Run Command output, and the script compared against that blank line. I fixed the script and re-ran the whole migration so the published numbers come from a clean run.

Part 3. Disaster recovery: pilot light in Ireland

Nothing runs in Ireland day to day. Two things are copied there continuously:

- **The database:** RDS cross-region automated backup replication sends snapshots and transaction logs to Ireland.
- **The files and the app release:** S3 replication copies every object to the Ireland bucket.

The same Terraform module that builds London builds Ireland. Recovery is one variable, `dr_active = true`, which restores the database to the latest restorable time and starts an app server pointed at the replica bucket.

Strategy	Runs in the DR region	Recovery time	Cost while idle
Backup and restore	Backups only	Hours	Lowest
Pilot light (this project)	Replicated data; servers built on demand	Minutes	Storage and transfer
Warm standby	A smaller copy, always running	Minutes or less	A second, smaller site
Active-active	A full copy serving traffic	Near zero	Two full sites

The drill

`scripts/dr-drill.sh` writes to London every five seconds for ten minutes, stops London's app server, turns on Ireland, and measures.

Measured	Result
Recovery time (RTO)	18 min 27 s, from outage to the app serving in Ireland
Data loss (RPO)	More than 10 minutes: none of the drill's 106 writes survived
Migrated data	All 501 jobs recovered
Replication lag	Ireland's latest restorable time was about 13.5 minutes behind London

What I took from it. Backup replication fits a business that can lose about 15 minutes of data, and most of the recovery time is the database restore. If the business needs seconds, the next step is a cross-region read replica: it stays seconds behind and is promoted rather than restored, at the cost of a second database running all the time.

Part 4. Testing and safety

Check	What it proves
App tests (pytest)	Filename sanitising, path traversal blocked, input validation, file storage round trip
Company server in CI	GitHub Actions builds the Docker Compose stack, seeds 500 jobs, uploads a file and reads it back
`terraform test`, mocked AWS	Ireland runs nothing until failover; backups replicate even while it is off; database private, encrypted, password generated; DR restores to the latest time; app never opened to the whole internet; backups cannot be switched off
Checkov	No security misconfigurations; each deliberate exception documented with its reason

Cost. Free-tier sizes, no NAT gateway, VPN or load balancer, and everything destroyed after the run, including the replicated backups AWS keeps after a database is deleted.

Part 5. Interview questions

"Why not AWS DMS or Application Migration Service?"

The database was small and the business could stop writing for a couple of minutes, so a native dump and restore was the simplest method I could verify end to end. DMS is right when the database is large or cannot stop, because it keeps both sides in sync until cutover. MGN is right for lifting many servers unchanged. Both bill for the replication servers they run.

"How do you know nothing was lost?"

I don't rely on row counts alone. I hash every row on both sides with the time zone fixed, and compare every file's SHA-256 with the checksum S3 recorded on upload. The script only reports success when all of them match.

"What are RTO and RPO, and what did you get?"

RTO is how long until the service is back; RPO is how much recent data you lose. My drill measured 18 and a half minutes and more than ten minutes. The RPO came from backup replication lag, which is why a read replica is the first upgrade for a business that needs better.

"How does the app get its database password?"

RDS generates it and stores it in Secrets Manager, which rotates it. The app server's IAM role can read that one secret, and the app fetches it when it connects, so a rotation needs no redeploy.

"How would users find the app after failover?"

With a domain, a Route 53 failover record with a health check moves users automatically. I documented it rather than deploying it because a hosted zone has a monthly charge and needs a domain.

Checks done (16 October 2026)

Check	Result
App tests	8 of 8 pass
terraform test	6 of 6 pass
Checkov	180 passed, 0 failed, 13 documented exceptions
Migration	Validated: 501 rows and 61 files matched; 94 s of frozen writes
DR drill	RTO 18 min 27 s; RPO more than 10 min
Teardown	All resources destroyed in both regions, including retained replicated backups